

American Thoracic Society (ATS), Centers for Disease Control and Prevention (CDC), European Respiratory Society (ERS), and Infectious Diseases Society of America (IDSA)

Updates on the Treatment of Drug-susceptible and Drug-Resistant Tuberculosis:

An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline

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11.6 Treatment of Tuberculosis or Latent Infection

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Recommendations Overview

Treatment of isoniazid-susceptible, rifampin-susceptible TB in adults with a 4-month rifapentine-moxifloxacin vs. 6-month regimen

Question: In adolescents and adults with drug-susceptible pulmonary TB, is a four month regimen composed of two months of isoniazid, rifapentine, pyrazinamide, and moxifloxacin followed by two months of isoniazid, rifapentine, and moxifloxacin (2HPZM/2HPM) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, and ethambutol (2HRZE) followed by four months of isoniazid, and rifampin (4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

Recommendation: In people aged 12 years or older with drug-susceptible pulmonary tuberculosis, we conditionally recommend the use of a 4-month regimen of isoniazid, rifapentine, moxifloxacin and pyrazinamide (Conditional recommendation, moderate certainty evidence). See Table 4 for dosing details.

Treatment of non-severe, presumed isoniazid- and rifampin-susceptible TB in children with 4-months vs. 6-months of standard therapy

Question: In children and adolescents with non-severe, drug-susceptible pulmonary TB, is a four month regimen composed of standard-dose two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by two months of isoniazid, rifampin (2HRZE/2HR) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by four months of isoniazid, rifampin (2HRZE/4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

Recommendation: In children and adolescents between three months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB), we recommend the use of a four-month treatment regimen 2HRZ(E)/2HR rather than the six-month drug-susceptible TB regimen of 2HRZ(E)/4HR. (Strong recommendation, moderate certainty of evidence)

Remarks: Non-severe TB is defined as peripheral lymph node TB; intrathoracic lymph node TB without airway obstruction; uncomplicated TB pleural effusion, or paucibacillary and non-cavitary disease confined to one lobe of the lungs, or without a miliary pattern. Children and adolescents who do not meet the criteria for non-severe TB should receive the standard 6-month treatment regimen (2HRZE/4HR), or recommended treatment regimens for severe forms of extrapulmonary TB. Dosing is found in Table 4.

Treatment of rifampin-resistant, fluoroquinolone-resistant TB with a 6-month BPaL regimen vs. 15-month or longer regimens in adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB

Question: In adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB is a six-month regimen composed of bedaquiline, pretomanid and linezolid as efficacious and safe as the current 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

Recommendation: In adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB with resistance or patient intolerance to fluoroquinolones, who have either had no previous exposure to bedaquiline and linezolid or have been exposed for less than 1 month we recommend the use of the 6-month treatment regimen, composed of bedaquiline, pretomanid and linezolid (BPaL), rather than more than 15-month regimens. (Strong recommendation, very low certainty of evidence). See Table 4 for dosing details.

Treatment of rifampin-resistant, fluoroquinolone-susceptible TB with a 6-month BPaLM regimen vs. 15-month or longer regimens in adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB

Question: In adolescents, aged 14 and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB is a six-month regimen composed of bedaquiline, pretomanid, linezolid and moxifloxacin as effective and safe as the 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

Recommendation: In adolescents, aged 14 and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB we recommend the use of a 6-month treatment regimen, composed of bedaquiline, pretomanid, linezolid, and moxifloxacin (BPaLM), rather than the 15-month or longer regimens in MDR/RR-TB patients. (Strong recommendation, very low certainty of evidence) See Table 4 for dosing details.

Abstract

Background Based on recent clinical trial data for treatment of drug-susceptible (DS) and drug-resistant (DR) tuberculosis (TB), the American Thoracic Society (ATS), U.S. Centers for Disease Control and Prevention (CDC), European Respiratory Society (ERS) and Infectious Diseases Society of America (IDSA) have updated clinical practice guidelines for TB treatment in children and adults in settings in which mycobacterial cultures, molecular and phenotypic drug susceptibility tests (DST), and radiographic studies, among other diagnostic tools, are available on a routine basis.

Methods A Joint Panel representing multiple interdisciplinary perspectives convened with ATS methodologists to review evidence and make recommendations using the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) and GRADE-ADOLOPMENT methodology.

Results New DS-TB recommendations include use of a novel 4-month regimen for people with pulmonary TB and a shortened 4-month regimen for children with non-severe TB. DR-TB recommendation updates include use of novel regimens containing bedaquiline, pretomanid, and linezolid with or without moxifloxacin.

Conclusions All-oral, shorter treatment regimens for TB are now recommended for use in eligible individuals.

Keywords tuberculosis, drug-resistant, drug-susceptible, children, adolescents, adults

Introduction

Successful treatment and cure of tuberculosis (TB) improves individual health and reduces *Mycobacterium tuberculosis* transmission. Historically, treatment of TB, particularly of drug-resistant TB (DR-TB), has involved prolonged courses of multiple medications and frequent adverse reactions. Patients and clinicians prefer shorter, safer, and more effective regimens with fewer pills and injections for drug-susceptible (DS) and DR-TB (Table 1). Recent clinical treatment trials of both DS- and DR-TB provided new evidence toward these objectives. Based on favorable study outcomes, the U.S. Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) updated DS- and DR-TB treatment recommendations in 2022 (1-4).

This guideline update, written by a panel of American Thoracic Society (ATS), CDC, European Respiratory Society (ERS), and Infectious Diseases Society of America (IDSA) specialists, relied on recent reviews of clinical data conducted by the WHO Guideline Development Group (GDG). This Joint Panel guideline focuses on TB in low-incidence settings without significant resource limitations.

Methods

Panel Composition

The Joint Panel was composed of 25 international specialists in pulmonary medicine, infectious diseases, pediatrics, epidemiology, and public health. A member of *We Are TB*, a patient advocacy organization, was included for patient perspective. All members helped to review and rate evidence. The co-sponsoring organizations each provided a co-chair and representatives responsible for the content of the manuscript.

Conflict-of-Interest Management

Guideline panel members disclosed all potential conflicts of interest (Col) according to the ATS policies (see Financial Disclosures). The Chairs and ATS reviewed and managed potential Cols. Panel members with potential Col abstained from decisions about specific questions and recommendations related to their potential Col. Most panelists had no substantial Col and were approved to participate without limitation. One protocol chair panelist (PN) participated in the discussions but was recused from formulating, grading, writing, or editing the recommendation related to the treatment of DS-TB in adults. Another panelist (AGP) had a manageable conflict requiring recusal from the formulation of antibiotic-related recommendations pertaining to DR-TB.

Guideline Development Methodology

This guideline is a targeted update to the prior Joint Panel's clinical practice guidelines on the treatment of TB (5) and specifically based as an adaptation of the WHO 2022 guidelines (1, 2) relevant to the prior Joint Panel's guideline recommendations (6, 7). Four methodologists followed the GRADE-ADOLOPMENT (1-4) process and used the Evidence to Decision (EtD) frameworks in the adaptation of published WHO guidelines (1, 2). ADOLOPMENT is a GRADE framework that involves either adoption of

recommendations from existing guidelines, adaptation of recommendations from existing guidelines, or if needed, de-novo development of recommendations (5).

The methodologists administered online surveys to confirm the panelists' selection of PICO (Population, Intervention, Comparison, Outcome) questions for ADOLOPMENT and prioritization of critical outcomes, based on recently published treatment trials and recent CDC and WHO guidelines (1-4) through a modified Delphi process. The Joint Panel arrived at consensus to use the WHO prioritization of critical outcomes. Specifically, the Joint Panel consensus, was to adapt specific questions on drug-sensitive and drug resistant TB as worded in the WHO guidelines pertaining to TB treatment (1, 2) provide an update to the 2016 and 2019 Joint Panel's Guidelines on TB treatment (6, 7) Subsequently the Joint Panel used the same wording of questions as noted in WHO guidelines, except to refer to the standard of care (SoC) noted in prior Joint Panel's guidelines as active comparators as opposed to active comparators from SoC from prior WHO guidelines (1-4).

The Joint Panel chose through consensus to adapt the 2022 WHO guidelines on TB treatment (1-4) based on the relevance, rigor, quality of evidence, and the GRADE methodology used in their development. Additionally, the methodology team conducted a targeted literature search in three databases (Medline, Embase, Cochrane Central Register of Clinical Trials), using "tuberculosis" and randomized clinical trial filter to confirm the absence of new clinical trial data pertinent to selected PICO questions since the publication of the WHO Guidelines in 2022_ (1, 2).

Literature Search

The followings steps were conducted after the literature search: i) titles and abstract screening (a preliminary screen review of all titles and abstracts to determine which studies are potentially eligible) ii) full-text screening (a more in-depth review of the full text for all included studies from the title/abstract phase to establish final eligibility for the review). For steps i) and ii) the methodology team screened the titles and abstracts (TA) of the uncovered citations and excluded studies on the basis of the pre-defined study selection criteria that was specific to each of the four study PICOs. Only studies that were deemed eligible (or were judged 'unsure' by screeners) went on to the full-text (FT) phase. At the TA and FT phase of screening, screening was performed in duplicate and independently by pairs of screeners with consensus debate to settle disagreements and 3rd party adjudication when/if needed. At the FT phase, the full studies were retrieved so that the screeners could read the study in entirety to make a definitive decision on whether a study would be retained to inform the respective PICO. Panel members were asked to be alert to any recently published trials or any studies that they thought were relevant to a particular PICO and that may have been missed in the electronic database search or TA and FT screen steps.

Evidence to Decision Tables

The methodology team provided transcribed versions of the EtDs of the selected WHO guidelines (1, 2) in GDT (<http://www.gradepro.org/>) (8). The terms, definitions of data, and evidence tables included in each EtD framework are derived from the published WHO consolidated guidelines on TB (1, 2) with the addition of data for the percentages of participants for each outcome and the treatment effect estimates derived from the evidence tables and EtD tables, respectively. The analyses documented in the published WHO guidelines EtD tables were used for the GRADE ADOLOPMENT process with inclusion of intervention and control population numbers from corresponding evidence tables. Joint Panel

members without a Col, reviewed the EtD tables derived from WHO (1, 2) EtD tables from corresponding PICO questions, and recommendations, and voted on recommendations. As part of the ADOLOPMENT process, the Joint Panel critically appraised and evaluated the WHO guidelines on TB treatment, particularly focusing on the relevance of the EtD judgments in the context of low TB burden, high-resource settings, in contrast to the WHO's focus on higher TB burden, low-resource settings (reassessed EtDs in supplement, tables S1-S4). The Joint Panel and methodology team revised the EtD tables transcribed from the WHO EtDs (1, 2) to reflect the Joint Panel reassessments of the judgements for each PICO. Recommendations were formulated and rated as either "strong" or "conditional" as explicit statements of recommendations strength as per GRADE guidance (Table 2)(9, 10). The term "we recommend" was used for a "strong" recommendation ; "we conditionally recommend" is used for a "conditional" recommendation, analogous to "we suggest" used in the Joint Panel's 2016 and 2019 guidelines (6, 7). The Joint Panel's recommendations are based on the quality of evidence, the balance between benefits and harms, the certainty of the evidence, and other factors as noted in Table 1. Sequential observational evidence was considered where published RCT evidence was not available. All panel members without a declared COI reviewed the EtD tables and recommendations and voted on the Joint Panel's recommendations. The Joint Panel and methodology team drafted the adapted guidelines EtDs and manuscript document.

Independent Review

The resulting final guideline was reviewed by the ATS Documents Editor, by anonymous peer reviewers, ERS, IDSA, and CDC, and approved for publication by the ATS Executive Committee.

Funding and updating

The ATS provided logistical and methodological support and was responsible for Col disclosures, vetting, and management. The guideline will be reevaluated for updating by the sponsoring ATS Assembly three years after publication or sooner.

Recommendations

Drug-Sensitive TB

PICO Question 1: In adolescents and adults with drug-susceptible pulmonary TB, is a four month regimen composed of two months of isoniazid, rifapentine, pyrazinamide, moxifloxacin followed by two months of isoniazid, rifapentine, moxifloxacin (2HPZM/2HPM) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, ethambutol (2HRZE) followed by four months of isoniazid, rifampin (4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

1. Topic/Overview/Background

Standard treatment for culture-positive DS-TB has required ≥ 6 months of antibiotics (6). Shorter, effective regimens enable patients to be cured faster and can potentially reduce treatment costs,

improve patient quality of life, and increase completion of therapy. The Joint Panel reviewed evidence from Study 31/A5349, a randomized, open-label, phase 3 trial comparing two 4-month rifapentine-containing regimens to the standard 6-month control regimen (11). Participants aged 12 years and older with newly diagnosed pulmonary TB were enrolled at 34 trial sites worldwide (Table 5).

2. Summary of evidence for benefits and harms

For the 4-month rifapentine and moxifloxacin (RPT-MOX) based regimen compared to the standard 6-month regimen (consisting of 2 months of isoniazid, rifampin, pyrazinamide, and ethambutol followed by 4-months of isoniazid and rifampin, 2HRZE/4HR), the Joint Panel assessed equivalent outcomes of TB disease-free survival at 12 months post-randomization (cure/favorable outcome); treatment retention; acquired drug resistance; adverse events (grade 3 or higher); and, all-cause mortality within 14 days after the last dose of study medication (Table 5, Table S1). The microbiologically eligible population for analysis had 791 participants in the 4-month RPT-MOX arm and 768 participants in the 6-month control arm, all with culture-confirmed susceptibility to isoniazid, rifampin, and fluoroquinolones. RPT-MOX was non-inferior to control (84.6% vs. 85.4% cure, respectively). No significant differences in outcomes were identified for RPT-MOX and 6-month standard regimens based on smear grade, cavitation, radiologic extent, age, diabetes, body weight, or HIV status. However, the number of participants living with HIV infection, diabetes, or extreme weight or age was small (e.g., around 9% of trial participants were people living with HIV infection with CD4 count ≥ 100 cells/ μ L, and most participants were aged under 35 years (11, 12) (Table S1). As part of the ADOLOPMENT process, we adapted the WHO guideline question, evidence table, EtD framework and recommendation to the scope and audience of this guideline. We performed a literature search to update the evidence base. The literature search revealed no other clinical trials to include in this question, and only one study and corresponding analysis from the WHO guideline was utilized. While other clinical trials included shorter regimens for the treatment of DS-TB, they did not address the specific PICO question regarding a rifapentine-moxifloxacin regimen.

3. Monitoring and additional considerations

Individuals with TB affecting the central nervous system, bones or joints, miliary tuberculosis and/or pericardial tuberculosis were not included in the above study and would not usually be treated with a 6- or 4-month regimen. In low incidence settings, more people with TB may be older than in the clinical trial and may take concurrent medications or have underlying comorbidities with higher risk for adverse events. Specifically, this may include fluoroquinolone-related adverse events, including neuropathology, tendinitis, QT prolongation, and dysglycemia (13). There was no evidence in the trial to suggest routine baseline or subsequent electrocardiographic (ECG) monitoring is required. The panel did not recommend baseline ECG monitoring for those receiving the shorter regimen (unless clinically indicated, such as by older age, presence of cardiac conditions, history of prolonged QT interval, or use of additional QT prolonging medication). The Joint Panel noted that the RPT-MOX regimen avoids the potential ocular toxicity of ethambutol and decreases the time to culture-conversion. Implementation of the RPT-MOX regimen may face challenges related to feasibility and cost. While the shorter treatment duration may benefit patients and healthcare systems, the RPT-MOX regimen increases daily pill-burden, especially compared with fixed-dose combinations (FDC) that might be used in some European locations for the 6-month regimen. The trial used directly observed therapy (DOT) for at least 5 of 7 treatment days a week as recommended (6), which may have improved adherence. DOT may not be feasible in settings that rely

on unsupervised self-administered therapy. Although the likelihood of fluoroquinolone resistance was considered small (<5%) in otherwise DS-TB, there is widespread community use of fluoroquinolones. Therefore, obtaining samples to send for molecular and possibly phenotypic fluoroquinolone DST is advisable prior to starting the RPT-MOX regimen. Availability and affordability of rifapentine, and other medications, pose significant challenges in some settings. Nitrosamine impurities in rifampin and rifapentine may also impact availability, though Joint Panel opinion agreed with the United States Food and Drug Administration assessment that the benefit of treating TB with rifamycins likely exceeds the risk (41). ECG monitoring might be used in some patients at risk for cardiac events and could increase costs. In terms of health equity, no information was available regarding the use of RPT-MOX for extrapulmonary TB (EPTB), or during pregnancy or in young children. Rifapentine with isoniazid was not associated with unfavorable pregnancy outcomes in trials including weekly therapy for latent tuberculosis infection treatment (14). Pediatric studies are enrolling for pharmacokinetics underlying the 4-month RPT-MOX regimen (ClinicalTrials.gov ID NCT03730181) and weekly rifapentine with isoniazid is recommended therapy for latent tuberculosis infection for persons aged 2 years and older (15).

4. Certainty of evidence

The Joint Panel concurred with the WHO GDG that overall certainty of evidence for benefits of cure and retention in treatment is high, moderate for adverse events, and low for acquisition of drug resistance and for all-cause mortality, based on low frequency of harmful events. While typically overall certainty is based on the lowest certainty for the agreed critical outcomes, the panel agreed with WHO in using GRADE guidance (10) that drug resistance and all-cause mortality would no longer rate as critical outcomes since the range of effect would not impact the strength or direction of recommendations. Thus, the panel concurred with WHO that the overall certainty of evidence is moderate.

5. Panel recommendation

In people aged 12 years or older with drug susceptible pulmonary tuberculosis, we conditionally recommend the use of a 4-month regimen of isoniazid, rifapentine, moxifloxacin and pyrazinamide (Conditional recommendation, moderate certainty evidence, Table S1). See Table 4 for dosing details.

PICO Question 2: In children and adolescents with non-severe, drug-susceptible pulmonary TB, is a four month regimen composed of standard-dose two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by two months of isoniazid, rifampin (2HRZE/2HR) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by four months of isoniazid, rifampin (2HRZE/4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

1. Topic/Overview/Background

Currently recommended treatment regimens are ≥6 months in duration, regardless of disease severity (6). The *Shorter treatment for minimal TB in children (SHINE)* trial (16) was a multicenter randomized, controlled two-arm non-inferiority trial comparing 4 months with 6 months of standard treatment for non-severe TB. This was defined as intrathoracic lymph node TB without airway obstruction;

uncomplicated TB pleural effusion, or paucibacillary and non-cavitary disease confined to one lobe of the lungs, or without a miliary pattern. Data from this clinical trial informed these recommendations (16). As part of the ADOLOPMENT process, we adapted the WHO guideline question, evidence table, EtD framework and recommendation to the scope and audience of this guideline. We performed a literature search to update the evidence base. The literature search revealed no other clinical trials to include in this question, and only the one study and corresponding analysis from the WHO guideline was utilized.

2. Summary of evidence, benefits, and harms

SHINE enrolled 1,204 children and adolescents <16 years old weighing ≥ 3 kg with non-severe TB disease (Table 6). Exclusion criteria included known drug resistance, contact to persons with DR-TB, pregnancy, and more severe TB disease (e.g., miliary TB, TB meningitis).

The SHINE trial demonstrated a success rate of 97.1% for participants receiving the 4-month regimen (2 months of isoniazid, rifampin, and pyrazinamide, with or without ethambutol, followed by 2 months of isoniazid and rifampin, 2HRZ(E)/2HR), compared to 96.9% among those receiving the 6-month regimen (2HRZ(E)/4HR, Table 6). Noninferiority of the 4-month regimen compared to the 6-month regimen was consistent across the intention-to-treat, per-protocol, and key secondary analyses, including the analyses restricted to patients independently adjudicated to have TB disease. Adverse event incidence was similar between the 4-month (7.8%) and the 6-month regimen (8.0%, Table 6).

The SHINE trial documented similar treatment success rates and adverse events between the 4-month and 6-month regimens, supporting a recommendation for the 4-month regimen. The shortened treatment duration is a significant benefit and could reduce use of healthcare services and potentially improve adherence.

Relying on the non-inferior outcome between the 4-month regimen and the current standard, the Panel judged both desirable and undesirable effects to be trivial in difference (2). The Joint Panel agreed that implementing the regimen would be cost saving for programs, though the size of this effect is uncertain.

3. Monitoring and additional considerations

While the SHINE trial enrolled children and adolescents with symptomatic TB disease, many children are diagnosed during contact investigations with TB disease while asymptomatic. The Joint Panel agreed that, provided patients meet the demographic, radiographic, epidemiologic, and clinical criteria for non-severe TB, without suspicion of RR/MDR-TB, these patients should also be eligible for the 4-month regimen. Recommendations for other specific subgroups are provided in Table 7.

Since there is an age overlap between the RPT/MOX regimen and this 4-month regimen, some children may be eligible for either regimen. Figure 1 provides information on how to identify children eligible for the 4-month regimens, based on the data on children in the SHINE trial (16) and Study 31/A5349 (11).

Children and adolescents who do not meet the criteria for non-severe TB should receive the standard 6-month treatment regimen (2HRZ(E)/4HR) or other recommended regimens for which they are eligible or for severe forms of EPTB (6).

The clinical monitoring requirements for the shorter regimen remain the same as for the 6-month regimen (6). Monitoring for potential recurrence is a priority for shorter regimens, and programs should have plans to monitor children post-treatment. TB programs should also assess for barriers to offering this regimen.

4. **Certainty of evidence**

The Joint Panel concurred with the WHO GDG that certainty in the estimated effect was judged to be high for treatment success. Specifically, the Joint Panel judged the certainty of the evidence to be high for treatment success and moderate for both all-cause death and adverse events. This was due to imprecision for estimates of treatment failure, relapse, adherence, and loss to follow-up. The Joint Panel agreed with WHO that overall certainty of the evidence is moderate (2). This was not downgraded for indirectness because the trial population may be representative of TB patients treated in the countries represented by the Joint Panel. The Joint Panel concurred with the WHO GDG that the certainty in the estimated effects was judged to be moderate for all-cause death and adverse events. The Joint Panel agreed with WHO that the overall certainty of evidence is moderate (2).

5. **Panel Recommendation**

In children and adolescents between 3 months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB), we recommend the use of a 4-month treatment regimen 2HRZ(E)/2HR rather than the 6-month drug-susceptible TB regimen of 2HRZ(E)/4HR. (Strong recommendation, moderate certainty of evidence, Table S2). See Table 4 for dosing details.

DRUG-RESISTANT TB

PICO Question 3: In adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB is a six-month regimen composed of bedaquiline, pretomanid, and linezolid as efficacious and safe as the current 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

1. **Topic/Overview/Background**

Multi-DR (MDR)-TB (resistant to at least isoniazid and rifampin), rifampin-resistant TB (RR-TB), and TB in people intolerant of rifampin (due to severity of adverse effects) are associated with significant morbidity and mortality. These conditions have required substantially different medications and extended treatment durations compared to DS-TB. Before 2019, global treatment success rates for MDR-TB ranged from 50 – 88% (1, 17) often accompanied by significant adverse events. However, the Nix-TB study demonstrated a 90% treatment success rate for treatment of extensively drug-resistant (XDR) or MDR-TB utilizing a regimen of bedaquiline, pretomanid, and linezolid (BPaL) over a 6-month period. Adverse events associated with this regimen were primarily attributed to a daily dose of 1200 mg of linezolid. In

the ZeNix Trial (18) lower linezolid dosing and shorter treatment durations were evaluated. WHO compared ZeNix trial data with programmatic individual patient data (IPD) of standard of care (SoC) regimens (Table S3).

WHO also analyzed data from TB-PRACTECAL (19), a randomized controlled trial, comparing programmatic SoC regimens with 6-month BPAL-based regimens (Table 8) for MDR-/RR- or pre-XDR TB. Data from these analyses were considered in this guideline. As part of the ADOLOPMENT process, we adapted the WHO guideline question, evidence table, evidence to decision framework and recommendation to the scope and audience of this guideline. We performed a literature search to update the evidence base. The literature search revealed no other clinical trials to include in this question, in addition to the clinical trials and corresponding evidence analysis noted in the WHO guideline.

2. Summary of evidence, benefits, and harms

In the ZeNix trial, 181 participants with MDR/RR- or pre-XDR-TB received 26 weeks of BPAL (Table 8). They were randomized to receive 1200 or 600 mg of linezolid daily for 9 or 26 weeks. Treatment with 600 mg of linezolid in the regimen for 26 weeks demonstrated similar favorable outcomes with a reduced incidence of peripheral neuropathy compared to the regimen with 1200 mg dosing. Comparing ZeNix data (linezolid 600mg for 26 weeks, n=43) with longer observational SoC IPD (n=850), BPAL had higher treatment success (100% vs 74%), with lower mortality (0% vs. 11%) and loss to follow-up (0% vs. 12%, Table 8). Grade 3 or higher adverse events were observed more frequently with BPAL (14% vs. 5%). Potential reporting bias in observational data might have led to underestimating adverse events in IPD. In TB-PRACTECAL (19), people taking BPAL (n=60) for 24 weeks, including linezolid initiated at 600 mg for MDR-/RR- or pre-XDR TB had a higher success rate than persons taking SoC (n=66) regimens (77% vs. 52%), and fewer grade 3 to 5 adverse events (20% vs. 51%, Table 8, Table S3).

3. Certainty of evidence

Certainty in estimated effects was very low due to multiple factors: bias risk from imbalance of comorbidities, high risk of unmeasured confounding, small event numbers, lack of blinding, early trial termination, population differences, varied comparator regimens (inclusion of nine-to-twenty-month regimens), inconsistent treatment outcomes, and imprecision from small participant and outcome numbers (Table S4). The ZeNix vs IPD-2021 analysis lacked adjustment due to few events, potential misclassification bias in the comparator group under programmatic conditions, and indirectness from assessment and procedural differences between intervention and programmatic comparators (1).

4. Monitoring and additional considerations

a. Patient selection

Patients should have MDR-/RR-TB confirmed by genotypic (e.g., Xpert MTB/Rif (original or Ultra), DNA sequencing, or line probe assay), or phenotypic methods. For patients with conditions excluded from

the above trials (Table 8), the risks, limited safety data, and benefits of BPaL or a more extended regimen should be carefully weighed and discussed with the patient before initiating therapy.

Individuals who have experienced rifampin intolerance would also be potentially eligible for BPaL, according to Joint Panel opinion and FDA. Although the minimum age for eligibility was 15 years in TB PRACTECAL and 14 years in ZeNix, the Joint Panel opinion was that BPaL could be used in individuals aged 14 years and older. Patients initiating treatment for MDR/RR-TB but ineligible for BPaL (e.g., resistance or intolerance to bedaquiline, pretomanid, or linezolid; severe extrapulmonary TB; pregnant; lactating; or aged < 14 years) can receive an individualized regimen based on the Joint Panel's 2019 DR-TB Guidelines (7). For individuals treated with BPaL who have received more than 9 weeks of linezolid, but are unable to tolerate it, expert opinion suggests that some individuals, after careful review, may continue the remaining medications without it (20). Prior recommendations discuss the role of surgery for DR-TB (7).

b. Subgroup considerations

Age

In low-incidence settings a greater proportion of people with TB may be older than in the ZeNix and TB PRACTECAL trials. Older individuals may take more concurrent medications or have comorbidities with a higher risk for adverse events.

Children, Pregnant, and Lactating Individuals

Trials included participants ≥ 14 years old, excluding pregnant or lactating individuals, with median participant age between 35 and 37 years old. Bedaquiline has a favorable pediatric adverse effect profile (21), while linezolid (22) and pretomanid data are limited or absent. At this time, longer regimens are recommended (7) for children under 14 years old and for lactating and pregnant individuals until safety and efficacy data become available.

People living with HIV_infection

People living with HIV infection comprised 20-33% of the ZeNix and TB PRACTECAL trial populations, with median CD4 counts of 326 and 330 cells/ μ L. However, subgroup outcome analyses were not offered in these trials. In the Nix-TB study, in which 51% of participants were living with HIV, unfavorable outcomes were not influenced by HIV status (23). Treatment recommendations for BPaL, based on Joint Panel opinion, may be applied to all patients with RR/MDR-TB regardless of HIV infection and CD4 count. The risks and benefits of 6-month or longer regimens should be discussed with the patient. Close monitoring is advised with low CD4 counts due to higher rates of disseminated disease and potential immune reconstitution syndrome. The potential for drug-drug interactions should be considered, in particular avoiding efavirenz in a BPaL –based regimen, as it can significantly decrease bedaquiline levels.

Extrapulmonary TB

Trials included patients with non-severe forms of EPTB, making such patients eligible for BPaL. Severe forms, such as disseminated central nervous system or bone/joint TB, were excluded. No clinical trial evidence supports their treatment with BPaL, though limited programmatic experience has been published (24).

c. Clinical monitoring

According to the Joint Panel opinion (Table 9), patients should be monitored during treatment for response to treatment and for adverse events, with clinical, radiologic, and laboratory assessments, including baseline, monthly, and as needed. Monitoring frequency should be increased for those of older age, people living with HIV, diabetes, baseline anemia, visual impairment, renal or liver disease, or if significant clinical, ECG, or laboratory abnormalities are found. Post-treatment monitoring for at least 1-2 years is recommended to identify recurrence.

Baseline and follow-up ECGs with QTc measurement are necessary during bedaquiline treatment. For BPaL without additional QT-prolonging agents, ECG should be conducted at 2, 12, and 24 weeks after initiation (25, 26). With a concomitant QT-prolonging agent, including moxifloxacin, as below, monthly ECGs are recommended (3).

d. Bacteriologic and Drug Susceptibility Testing (DST)

Sputum should be obtained before treatment for DST and monitored weekly until smear conversion, every other week until culture conversion, and then monthly until the end of treatment. Isolates should be evaluated for susceptibility changes. Molecular methods, available through some state and national laboratories, such as the CDC (3), can detect moxifloxacin resistance and should be followed, when possible, by phenotypic testing as some molecular methods may miss hetero-resistant populations (27, 28). DST is limited for bedaquiline, linezolid, and pretomanid, but may become more available. DNA sequencing can detect mutations in genetic loci associated with resistance to BPaL drugs (29-31). If resistance to a drug in BPaL is detected by a molecular method or DST, an alternate regimen should be considered in consultation with laboratory and clinical experts, tailored to susceptibility results (7).

e. Drug-drug interactions

Drug-drug interactions should always be considered. Classes of drugs and specific medications to avoid or that require additional precautions include, but are not limited to, efavirenz, QTc-prolonging medications, strong CYP3A4 inducers (including rifamycins), strong CYP3A4 inhibitors, monoamine oxidase inhibitors, many antidepressants, and drugs known to induce myelosuppression (Tables S3 & S4).

f. Regimen administration, composition, dosing, frequency, duration, and changes

DOT and integrated case management remain the standard of care, as noted in the Joint Panel's prior guidance (6, 7). The dosing of BPAL-based regimens is shown in Table 4. The Joint Panel acknowledges the two-week difference in treatment duration between ZeNix and TB-PRACTECAL and recommends a treatment duration of 26 weeks, considering expert opinion and CDC guidance (3). Early discontinuation or replacement of BPAL components may lead to worse treatment outcomes. In the Nix-TB and ZeNix trials, treatment was extended for a few patients to 39 weeks if sputum remained culture-positive or reverted to positive between months 4 and 6, or if clinical condition suggested progressive TB. For such patients or suspected failure, we recommend a complete re-evaluation, including both molecular and phenotypic DST testing for resistance to components of BPAL, and weighing the individualized risks and benefits of extending BPAL to 39 weeks. Although, at this time, there is insufficient clinical trial evidence for or against such prolongation, expert opinion suggests consideration of treatment extension to 39 weeks if there is delayed treatment response (e.g., culture conversion > 8 weeks with clinical condition slow to improve (3).

g. Linezolid dosing

The preferred dose of linezolid is 600 mg daily throughout the regimen. Therapeutic drug monitoring may help guide linezolid dosing with a goal trough of less than 2 mcg/mL to minimize adverse reactions (32). This may be available through some public health or academic TB laboratories. If the trough level is elevated, the Joint Panel suggests the dosing interval should be increased or the dose lowered.

5. Panel recommendation

In adolescents aged 14 years and older and adults with rifampin-resistant pulmonary TB who have resistance or patient intolerance to fluoroquinolones and have either had no previous exposure to bedaquiline and linezolid or have been exposed for less than 1 month, we recommend the use of the 6-month treatment regimen, composed of bedaquiline, pretomanid, and linezolid (BPAL), rather than more than 15-month regimens. (Strong recommendation, very low certainty of evidence, Table S3). See Table 4 for dosing details.

PICO Question 4: In adolescents, aged 14 and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB, is a six-month regimen composed of bedaquiline, pretomanid, linezolid, and moxifloxacin as effective and safe as the 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

1. Topic/Overview/Background

As discussed above, MDR-TB, RR-TB, and TB in individuals intolerant of rifampin have been associated with extended treatment duration and major morbidity and mortality. In the TB-PRACTECAL trial (19), , which included study participants with MDR/RR- or pre-XDR-TB with or without fluoroquinolone resistance, WHO standard of care (SoC) regimens were compared with three all-oral 24-week investigative regimens, including a BPaL plus moxifloxacin (BPaLM) arm. The SoC regimens included 9–12-month injectable-containing regimens; 18–24-month long pre-2019 WHO regimens; a 9–12 month all oral regimen; and an 18–20 month all oral regimen. The study was an all open-label, multicenter, randomized non-inferiority trial with a primary outcome of unfavorable status, which was a composite of death, treatment failure, treatment discontinuation, loss to follow-up, or recurrence of tuberculosis, at 72 weeks after randomization (19) . TB-PRACTECAL provided evidence for an assessment by WHO in 2022 (1). For individuals treated with BPaLM who have received more than 9 weeks of linezolid, but are unable to tolerate it, expert opinion suggests that some individuals, after careful review, may continue the remaining medications without it (20). Prior recommendations discuss the role of surgery for DR-TB (6).

2. Summary of evidence, benefits, and harms

Participants receiving the BPaLM regimen (n=62) compared to those treated with SoC regimens (n=66) experienced higher levels of treatment success (89% vs 52%), lower levels of failure and recurrence (8% vs 26%); lower levels of loss to follow-up (3% vs 20; and lower levels of grade 3 to 5 adverse events (21% vs 51%), see Table 8.

3. Certainty of evidence

Certainty in estimated effects was very low due to multiple factors: bias risk from imbalance of comorbidities, high risk of unmeasured confounding, small event numbers, lack of blinding, early trial termination, population differences, varied comparator regimens (inclusion of nine-to-twenty-month regimens), inconsistent treatment outcomes, and imprecision from small participant and outcome numbers (Table S4).

4, Monitoring and other considerations

a. Patient selection

The same selection considerations as in PICO 3 apply, but with the absence of known fluoroquinolone resistance here.

b. Subgroup considerations

Age

In low-incidence settings a greater proportion of people with TB may be older than in the ZeNix and TB PRACTECAL trials. Older individuals may take more concurrent medications or have comorbidities with a higher risk for adverse events, including fluoroquinolone-related adverse events, such as neuropathy, tendinitis, QT prolongation, and dysglycemia (13) .

Children, Pregnant, and Lactating Individuals

Trials included participants ≥ 14 years old, excluding pregnant or lactating individuals, with median participant age between 35 and 37 years old. Bedaquiline has a favorable pediatric adverse effect profile (21), while linezolid (22) and pretomanid data are limited or absent. At this time, longer regimens are recommended (6) for children under 14 years old and for lactating and pregnant individuals until safety and efficacy data become available. Moxifloxacin in the BPaLM regimen was well tolerated in the age groups enrolled in TB PRACTECAL(19) .

People living with HIV_infection

People living with HIV infection comprised 33% of the TB PRACTECAL trial population, with median CD4 counts of 326 and 330 cells/ μ L, though subgroup outcome analyses were not offered in these trials. In the Nix-TB study, in which 51% of participants were living with HIV, unfavorable outcomes were not influenced by HIV status (23). Treatment recommendations for BPaLM, based on expert opinion, may be applied to all patients with RR/MDR-TB regardless of HIV infection and CD4 count. The risks and benefits of 6-month or longer regimens should be discussed with the patient. Close monitoring is advised with low CD4 counts due to higher rates of disseminated disease and potential immune reconstitution syndrome. The potential for drug-drug interactions should be considered, in particular avoiding efavirenz in a BPaL –based regimen, as it can significantly decrease bedaquiline levels.

Extrapulmonary TB

Trials included patients with non-severe forms of EPTB, making such patients eligible for BPaLM. Severe forms, such as disseminated central nervous system or bone/joint TB, were excluded. No clinical trial evidence supports their treatment with BPaLM, though limited programmatic experience has been published (24).

c. Clinical evaluation, monitoring, and evaluation

During treatment, patients should be monitored for treatment response and adverse events, with clinical, radiologic and laboratory assessments, including baseline, monthly, and as needed, according to Joint Panel opinion (Table 9). Monitoring frequency should be increased for those of older age, people living with HIV, diabetes, baseline anemia, visual impairment, renal or liver disease, or if significant clinical or laboratory abnormalities are found. Post-treatment monitoring for at least 1-2 years is recommended to identify recurrence. During BPaLM treatment, baseline and follow-up monthly ECGs with QTc measurement are necessary (3, 25, 26).

d. Bacteriologic and Drug Susceptibility Testing (DST)

Sputum cultures should be obtained before treatment for DST and monitored weekly until smear conversion, every other week until culture conversion, and then monthly until the end of treatment. Isolates should be evaluated for susceptibility changes. Molecular methods, available through some state and national laboratories, such as the CDC (3), can detect moxifloxacin resistance and should be followed, when possible, by phenotypic testing as some molecular methods may miss hetero-resistant populations (27, 28). DST is limited for bedaquiline, linezolid, and pretomanid, but may become more available. DNA sequencing can detect mutations in genetic loci associated with resistance to BPALM drugs (29-31). If resistance to a drug in BPALM is detected by a molecular method or DST, an alternate regimen should be considered in consultation with laboratory and clinical experts tailored to susceptibility results (7).

e. Drug-drug interactions

Drug-drug interactions should always be considered. Classes and specific medications to avoid or that require additional precautions include, but are not limited to, efavirenz, QTc-prolonging medications, strong CYP3A4 inducers (including rifamycins), strong CYP3A4 inhibitors, monoamine oxidase inhibitors, many antidepressants, and drugs known to induce myelosuppression (Tables S3 and S4).

f. Regimen administration, composition, dosing, frequency, duration, and changes

DOT and integrated case management remain standard of care, as noted in the Joint Panel's 2019 guidance (6). The dosing of BPAL-based regimens is shown in Table 4. The Joint Panel acknowledges the two-week difference in treatment duration between ZeNix and TB-PRACTECAL and recommends a treatment duration of 26 weeks, considering expert opinion and CDC guidance (3). Early discontinuation or replacement of BPALM components may lead to worse treatment outcomes. In the Nix-TB and ZeNix trials, treatment was extended for a few patients to 39 weeks if sputum remained culture-positive or reverted to positive between months 4 and 6, or if clinical condition suggested progressive TB. For such patients or suspected treatment failure, we recommend a complete re-evaluation, including both molecular and phenotypic DST testing for resistance to components of BPALM, and weighing the individualized risks and benefits of extending BPALM to 39 weeks. Although, at this time, there is insufficient clinical trial evidence for or against such prolongation, expert opinion suggests consideration of treatment extension to 39 weeks if there is delayed treatment response (e.g., culture conversion > 8 weeks with clinical condition slow to improve) (3).

g. Linezolid dosing The preferred dose of linezolid is 600 mg daily throughout the regimen. Therapeutic drug monitoring may help guide linezolid dosing with a goal trough of less than 2 mcg/mL to minimize adverse reactions (32). If the trough level is elevated, Joint Panel opinion suggests the dosing interval should be increased or the dose lowered.

2. Panel recommendation

*In adolescents, aged 14 years and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB, **we recommend** the use of a 6-month treatment regimen, composed of bedaquiline, pretomanid, linezolid, and moxifloxacin (BPaLM), rather than the 15-month or longer regimens in MDR/RR-TB patients. (Strong recommendation, very low certainty of evidence, Table S4).*

Discussion

Shorter treatment duration, oral regimens, reduced number of medications, reduced pill burden, and less adverse drug effects are valued by patients and providers (35-37).

Individuals with DS-TB may be eligible for effective regimens that reduce treatment duration by a third. Previously, only patients with smear- and culture-negative "non-severe" pulmonary TB were recommended to receive 4 months of treatment (6). Adults and adolescents with DS-TB may now be offered 4 months of RPT-MOX-based regimen. Most children with non-severe TB can now be treated with 4 months of standard medications. For DR-TB, new all-oral 6-month regimens are more effective and safer than SoC regimens used for decades and may reduce morbidity and mortality. The BPaL-based regimens can be considered applicable also to those with intolerance of rifampin.

The regimens and staff training can be implemented within existing programmatic efforts. Access to regimens, monitoring tests and DST may vary and should be assured. These regimens may allow redirection of resources for other programmatic needs. Patient-centered support, case management, education, and, in some cases, incentives, may improve adherence and treatment success (6, 7).

The new shorter regimens are probably acceptable to clinicians and people with TB. The SHINE acceptability sub-study (16) indicated that treatment administration was more difficult for younger than older children but may be offset by an 8-week shorter regimen. In the United States, BPaL regimen uptake within two years of Food and Drug Administration (FDA) approval has been reported with good clinical outcomes (24).

Access to medications is variable, notably FDC's, rifapentine, bedaquiline, and pretomanid, which may improve. Child-friendly FDC's, which reduce pill burden, are available from the Stop TB Partnership's Global Drug Facility, but not in the US or many places in Europe. Increased availability would promote uptake of these recommended regimens. Net costs of implementing new regimens may improve with fewer patient visits, potentially offsetting higher costs of medications and DST. Studies assessing the cost-effectiveness, impact on health equity, acceptability, and feasibility of these newest regimens are needed.

Evidence supports higher treatment success rates for MDR/RR-TB with BPaL/BALM treatment, with fewer adverse events and shorter duration than with longer SoC regimens. The 600 mg linezolid dose is preferred. A study in a country with high TB burden (33) found that BPaL-based regimens are cost-saving and more effective than WHO SoC regimens. In low TB incidence countries, bedaquiline is currently expensive and can be difficult to access. The all-oral, shorter-duration regimen with a reduced pill

burden offered by BPaL/BPaLM is likely to confer substantial benefits, although these advantages were not directly quantified in clinical trials.

Based on the evidence available, the critical outcomes supported the benefits of BPaL/BPaLM. Although the certainty of evidence for adverse events was low, the likelihood of severe adverse events or amplification of resistance occurring was low.

A strong recommendation based upon very low-quality evidence is justified because the evidence that the recommended regimen causes less overall harm is high-quality (34) despite the low certainty of evidence for desirable outcomes. Specifically, the guideline committee had decided that the lone critical outcome for harms would be *on-treatment* adverse effects. This outcome was not measured in the published evidence, which had included follow-up beyond treatment completion. Therefore the Joint Panel considered unpublished observational evidence as sequential evidence (38). Joint Panel considered unpublished observational data from Panel members' clinical experience in the use of both BPaL and BPaLM. The unpublished observational evidence indicated that standard therapy was associated with a very large increase in on-treatment adverse effects, which constitutes high-quality evidence. While observationally based outcomes would, at first consideration, be rated as low certainty of evidence, with large treatment effect, the rating of the on-treatment adverse events as a critical outcome elevates by two levels the evidence to high quality of evidence.

The Joint Panel judged the 6- month BPaL/M regimens to be safer and easier for patient tolerance and adherence relative to prior longer SoC regimens. As the Joint Panel discussed while preparing this guideline, the *de facto* situation is that the 6-month all-oral BPaL regimens are the accepted treatment of choice in TB treatment centers in higher resourced countries that have access to BPaL. Other outcomes considered important to decision-making include the potential for medication adverse effects to be drawn out over long periods of time with the SoC regimens, which is not captured by the adverse event incidence data; the high value placed on the all-oral, shorter BPaL and BPaLM regimens exemplified by their widespread adoption; and the potential for more rapid reintegration of treated patients into society which may be related to more rapid de-stigmatization and improved health. The Joint Panel feels that there are significant harms associated with longer SoC regimens that are improved upon by the 6-month BPaL-based regimens. Thus, the strong recommendation in favor of the 6-month regimens is warranted.

We recommend BPaLM as the first line for MDR/RR-TB without fluoroquinolone resistance or intolerance. While BPaL is an alternative, the Joint Panel felt that BPaLM, with DOT and close monitoring, might safely add protection against the emergence of drug resistance and increase culture conversion at 2 months. The BPaL-based regimens can be considered to apply to those with intolerance of rifampin. Monitoring DST for the emergence of resistance during treatment is very important.

The main limitations of this guideline update are methodological, utilizing ADOLOPMENT, and the level of certainty of evidence. We did not review raw data available to WHO. For all of our PICO questions, we specifically focused upon WHO recommendations that updated prior Joint Panel guidance, and

several newer regimens were out of scope of the selected PICO questions. Our PICO questions 3 and 4 were focused on 15-month or greater comparator regimens, though the WHO data available included 9-month treatment regimens. The certainty of the evidence for some outcomes was very low. Nevertheless, the beneficial treatment effects, all-oral regimen, and relatively few adverse events favored the shorter intervention regimens.

These recommendations are based on single or small studies, which may not reflect TB populations in US and Europe. TB clinical trial participants are younger than TB patients in the US and Europe (39, 40). Adverse events may be more common in older individuals and may differ between trial and TB programmatic settings. These studies also excluded children with severe TB or DR-TB, pregnant individuals, and patients with severe forms of EPTB. Studies of treatment of individuals excluded from these trials, new regimens, emergence of drug resistance and other issues, will inform future guidelines (Table 10).

This Joint Panel Guideline updates the 2016 and 2019 TB Treatment Guidelines (6, 7) in specific types of DS- and DR-TB. For patients who do not fit these categories, the Joint Panel's prior guidelines should be consulted (6, 7). This document's focus is TB low-incidence high-resource contexts, in contrast to WHO's TB high-burden, lower-resource settings. DOT and integrated case management remain standard of care, as noted in the Joint Panel's prior guidance (6, 7). The Joint Panel offered a strong recommendation for BPAL/BPALM for MDR-/RR-TB whereas the WHO GDG provided a conditional recommendation. We continue to recommend DST to guide therapy and DOT as standard of care for TB treatment.

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Figure 1: Identifying children eligible for 4-month regimens

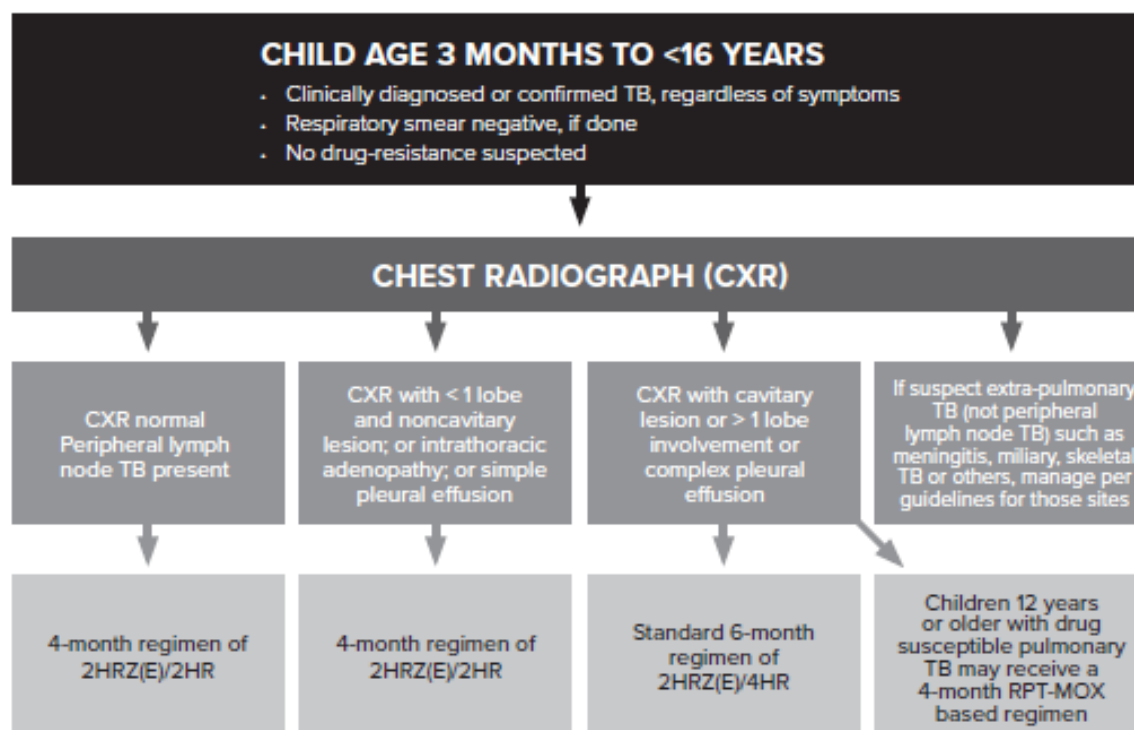


Figure developed by the Joint Panel, based on the data in the SHINE trial and Study 31/A5349, to address eligibility of some children for either the RPT/MOX regimen or the 4-month standard drug regimen.

Abbreviations

CXR: Chest radiograph

HRZE: Isoniazid, rifampin, pyrazinamide, Ethambutol

RPT-MOX: rifapentine-moxifloxacin

Table 1: Patient perspective

“For individuals living with tuberculosis, the path from diagnosis to recovery is full of many challenges. Delay in diagnosis and prolonged symptoms can be common. Long periods of isolation carry both economic and emotional costs. The difficulty of high pill burden, medication side effects and long treatment regimens is frequently discussed by TB survivors in *We Are TB* support meetings. During my own treatment I felt the weight of 16 pills in my hand every morning, and of stigma, financial cost and isolation. I felt the added burden on my family, the medication side effects, and the physical manifestations of the disease. Progress to shorten this journey and to ease these burdens is valued by patients.”

-- Kelly Holland, *We Are TB* patient advocate

Table 2. Implications of Strong and Conditional Recommendations

	Strong Recommendation ("We recommend . . .") *	Conditional Recommendation ("We conditionally recommend . . .")**
For Patients	The overwhelming majority of individuals in this situation would want the recommended course of action, and only a small minority would not.	The majority of individuals in this situation would want the suggested course of action, but a sizeable minority would not.
For Clinicians	The overwhelming majority of individuals should receive the recommended course of action. Adherence to this recommendation according to the guideline could be used as a quality criterion or performance indicator. Formal decision aids are not likely to be needed to help individuals make decisions consistent with their values and preferences.	Different choices will be appropriate for different patients, and you must help each patient arrive at a management decision consistent with her or his values and preferences. Decision aids may be useful to help individuals make decisions consistent with their values and preferences. Clinicians should expect to spend more time with patients when working toward a decision.
For Policy Makers	The recommendation can be adapted as policy in most situations, including for the use as performance indicators.	Policy making will require substantial debates and involvement of many stakeholders. Policies are also more likely to vary between regions. Performance indicators would have to focus on the fact that adequate deliberation about the management options has taken place.

* Strong recommendations will apply to most patients for whom these recommendations are made, but they may not apply to all patients in all conditions; no recommendation can take into account all of the unique features of individual patients and clinical circumstances.

** "We conditionally recommend" is analogous to the term "we suggest" in the 2016 and 2019 Joint Panel TB guidelines.

Table 3. PICO questions and recommendations regarding the treatment of tuberculosis selected by the Joint Panel

Treatment of isoniazid-susceptible, rifampin-susceptible TB in adults

Q1. In adolescents and adults with drug-susceptible pulmonary TB is a four month regimen composed of two months of isoniazid, rifapentine, pyrazinamide, moxifloxacin followed by two months of isoniazid, rifapentine, moxifloxacin (2HPZM/2HPM) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, ethambutol (2HRZE) followed by four months of isoniazid, rifampin (4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

R1. In people aged 12 years or older with drug-susceptible pulmonary tuberculosis, we conditionally recommend the use of a 4-month regimen of isoniazid, rifapentine, moxifloxacin and pyrazinamide (Conditional recommendation, moderate certainty evidence).

Treatment of non-severe, presumed isoniazid- and rifampin-susceptible TB in children

Q2. In children and adolescents with non-severe, drug-susceptible pulmonary TB is a four month regimen composed of standard-dose two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by two months of isoniazid, rifampin (2HRZE/2HR) as efficacious and safe as the standard six month drug-susceptible TB regimen of two months of isoniazid, rifampin, pyrazinamide, ethambutol followed by four months of isoniazid, rifampin (2HRZE/4HR) endorsed by the ATS/CDC/ERS/IDSA guidelines?

R2. In children and adolescents between three months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB), we recommend the use of a four-month treatment regimen 2HRZ(E)/2HR rather than the six-month drug-susceptible TB regimen of 2HRZ(E)/4HR. (Strong recommendation, moderate certainty of evidence)

Remarks: Non-severe TB is defined as peripheral lymph node TB; intrathoracic lymph node TB without airway obstruction; uncomplicated TB pleural effusion, or paucibacillary and non-cavitary disease confined to one lobe of the lungs, or without a miliary pattern. Children and adolescents who do not meet the criteria for non-severe TB should receive the standard 6-month treatment regimen (2HRZE/4HR), or recommended treatment regimens for severe forms of extrapulmonary TB.

Treatment of rifampin-resistant, fluoroquinolone-resistant TB

Q3. In adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB is a six-month regimen composed of bedaquiline, pretomanid and linezolid as efficacious and safe as the current 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

R3. In adolescents, aged 14 and older, and adults with rifampin-resistant pulmonary TB with resistance or patient intolerance to fluoroquinolones, who have either had no previous exposure to bedaquiline and linezolid or have been exposed for less than 1 month we recommend the use of the 6-month treatment regimen, composed of bedaquiline, pretomanid and linezolid (BPaL), rather than more than 15-month regimens. (Strong recommendation, very low certainty of evidence)

Treatment of rifampin-resistant, fluoroquinolone-susceptible TB

Q4. In adolescents, aged 14 and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB is a six-month regimen composed of bedaquiline, pretomanid, linezolid and moxifloxacin as effective and safe as the 15 months or longer drug-resistant TB regimens composed according to current ATS/CDC/ERS/IDSA DR-TB Treatment guidelines?

R4. In adolescents, aged 14 and older, and adults with rifampin-resistant, fluoroquinolone-susceptible pulmonary TB we recommend the use of a 6-month treatment regimen, composed of bedaquiline, pretomanid, linezolid, and

moxifloxacin (BPaLM), rather than the 15-month or longer regimens in MDR/RR-TB patients. (Strong recommendation, very low certainty of evidence)

Table 4: Recommended Drug Regimens

Q1: Treatment of isoniazid-susceptible, rifampin-susceptible TB in adults

Recommended 4-month rifapentine-moxifloxacin containing regimen*	
Isoniazid†	300 mg daily for 17 weeks
Rifapentine	1200 mg daily for 17 weeks
Pyrazinamide	Weight-based dosing daily for 8 weeks: 40–<55 kg: 1,000 mg ≥55–75 kg: 1500 mg >75 kg: 2000 mg
Moxifloxacin	400 mg daily for 17 weeks

*Using actual body weight. Medications should be administered 7 days/week with food, avoiding milk, antacids, or other cationic items, with DOT 5 of 7 days per week.

†Pyridoxine (vitamin B6), 25–50 mg/day, should be given with isoniazid to all patients.

Q2: Treatment of non-severe, presumed isoniazid-susceptible, rifampin-susceptible TB in children

Recommended Regimen	Intensive Phase, 8 weeks*	Continuation Phase, 8 weeks
Isoniazid	10–15 mg/kg	10–15 mg/kg
Rifampin	10–20 mg/kg	10–20 mg/kg
Pyrazinamide	35 (30–40) mg/kg	None
Ethambutol †	20 (15–25) mg/kg (included/excluded based on local guidelines)	None

* Using actual body weight and DOT 5 of 7 days per week.

† To avoid potential ocular toxicity, some clinicians exclude ethambutol for children who are HIV-uninfected, have no prior TB treatment history, live in an area of low prevalence of DR-TB, and have no exposure to an individual from an area of high prevalence of DR-TB. Prevalence and risk factors can be

difficult to ascertain, therefore, the American Academy of Pediatrics and most experts include ethambutol as part of the intensive-phase regimen for children with TB.

Q3: Treatment of rifampin-resistant, fluoroquinolone resistant TB

Recommended BPaL regimen*	
Bedaquiline	400mg daily for 2 weeks, then 200mg three times/week subsequent 24 weeks
Pretomanid	200mg daily for 26 weeks
Linezolid	600mg daily for 26 weeks

* Medications should be administered 7 days/week with food, with DOT 5 of 7 days per week.

Q4: Treatment of rifampin-resistant, fluoroquinolone sensitive TB

Recommended BPaLM regimen*	
Bedaquiline	400mg daily for 2 weeks, then 200mg three times/week for subsequent 24 weeks
Pretomanid	200mg daily for 26 weeks
Linezolid	600mg daily for 26 weeks
Moxifloxacin	400mg daily for 26 weeks

* Medications should be administered 7 days/week with food, avoiding milk, antacids, or other cationic items with DOT 5 of 7 days per week.

Table 5. Study 31/A5349 evidence: Drug regimens and WHO critical outcomes in the treatment of isoniazid-susceptible, rifampin-susceptible TB in adults*

Study	Study 31/A5349*
Design	-Randomized, open-label, phase 3, noninferiority trial -Compared a 4-month regimen (including moxifloxacin and rifapentine) with standard 6-month regimen
Setting	International, multicenter (34 sites): Brazil, China, Haiti, India, Kenya, Malawi, Peru, South Africa, Thailand, Uganda, United States of America, Vietnam, Zimbabwe)
Inclusion Criteria	-Participants ≥ 12 years old with TB (sputum AFB smear or rapid nucleic acid amplification test positive) susceptible to isoniazid, rifampin, and fluoroquinolones -If HIV positive, CD4 T cell count ≥ 100 cells/mm³, on (or planned) EFV based ART
Exclusion Criteria	-Pregnant and breastfeeding women -Receiving >5 days of treatment directed against TB or latent TB infection within prior 6 months or, >5 days treatment latent TB infection treatment with isoniazid, rifamycins, pyrazinamide or any fluoroquinolone within prior 30 days -Known history of prolonged QT syndrome -Extrapulmonary TB (CNS, bones or joints, miliary) -Weight <40 kg -Known drug resistance

Regimen and duration	Intervention: 4-month regimen 2HPZM/ 2HPM †	Comparator: 6-month regimen 2HRZE/ 4HR ‡		
WHO Outcomes	Intervention Percentage	Comparator Percentage	Relative effect (95% CI) Anticipated absolute difference (95% CI)	Certainty of the evidence (GRADE)
Cure n=1559	84.6%	85.4%	RR 0.99 (0.95 to 1.03) 9 fewer per 1,000 (from 43 fewer to 26 more)	⊕⊕⊕⊕ High
Retention in treatment n= 1559	99.9%	99.0%	RR 1.01 (1.00 to 1.02) 10 more per 1,000 (from 0 fewer to 20 more)	⊕⊕⊕⊕ High
Amplified drug resistance n= 1559	0.0%	0.0%	RR 3.13 (0.13 to 76.69) 0 fewer per 1,000 (from 0 fewer to 0 more)	⊕⊕○○○ Low
Adverse events during treatment (grade 3 or higher) n= 1671	18.7%	19.3%	RR 0.97 (0.76 to 1.24) 6 fewer per 1,000 (from 46 fewer to 46 more)	⊕⊕⊕○ Moderate
All-cause mortality (within 14 days after end of treatment) n= 1671	0.4%	0.8%	RR 0.42 (0.11 to 1.61) 5 fewer per 1,000 (from 8 fewer to 5 more)	⊕⊕○○○ Low

* Adapted from WHO DS 2022 TB treatment guidelines Evidence to Decision tables (From Module 4 Treatment, Web annex EtD corresponding to question “ In patients aged ≥ 12 years with drug-susceptible pulmonary TB is a 4 month regimen composed of rifapentine, isoniazid, pyrazinamide and moxifloxacin as effective and safe as the standard drug-susceptible TB regimen composed according to WHO guidelines?”, pages 20-41)

† Four months total with two months of Isoniazid, Rifapentine, Pyrazinamide, Moxifloxacin (2HPZM) followed by two months Isoniazid, Rifapentine, Moxifloxacin (2HPM)

‡ Six months total with two months of Isoniazid, Rifampin, Pyrazinamide, Ethambutol (2HRZE) followed by four months of Isoniazid, Rifampin (4HR)

Table 6. SHINE trial evidence: Drug regimens and WHO analyzed critical outcomes in the treatment of non-severe presumed isoniazid-susceptible, rifampin-susceptible TB in children*

Study	SHINE trial *
Design	-Open-label, parallel-group, non-inferiority, randomized controlled, two-arm trial -Compared 4-month (16 weeks) versus standard 6-month (24 weeks) treatment using WHO-recommended anti-TB drug dosing in children <16 years with symptomatic non-severe TB. -Sample size: 1121 (per-protocol; PP), 1145 (modified intention-to-treat; mITT), 1204 (intention-to-treat; ITT)
Setting	Uganda, Zambia, South Africa, India
Inclusion Criteria	-Age 0–16 years -Weight ≥ 3 kg -Symptomatic, non-severe TB -Clinician decision to treat with standard first-line regimen
Exclusion Criteria	-Smear-positive respiratory sample -cavitation on chest radiograph -Premature (< 37 weeks) and aged under 3 months -Miliary TB, spinal TB, TB meningitis, osteoarticular TB, abdominal TB, congenital TB -Pre-existing non-tuberculous disease likely to prejudice the response to, or assessment of, treatment, e.g. liver or kidney disease, peripheral neuropathy, - -Any known contraindication to taking anti-TB drugs -Known contact with drug-resistant adult source case (including mono-resistant TB) or known drug resistance in the child -Severely ill -Pregnancy

Sympto- matic, non- severe TB defined		-Disease confined to one lobe on chest radiograph -No cavities or complex pleural effusion, or significant airway obstruction -Extra-thoracic lymph node (LN) TB -Intrathoracic uncomplicated (hilar) LN TB -Smear-negative on gastric aspirate/other respiratory sample			
		- HIV seropositive	11%		
		-respiratory TB	67%		
Participant characteris- tics	Median age 3.5	-peripheral LN TB	3%	Adherence to assigned treatment: 94%	
	years (range 2	-mixed respiratory and			
	months–15	peripheral LN TB	29%		
	years)				
	52% male	-bacteriologically confirmed TB	14%		
		-culture-positive	7%		
		-Xpert MTB/RIF-positive	2%		
		-culture-Xpert MTB/RIF-positive	5%		
Drug regimen and duration	Intervention: 4 months of isoniazid, rifampin, pyrazinamide with or without ethambutol†	Comparator: 6 months of isoniazid, rifampin, pyrazinamide with or without ethambutol ‡			
WHO Outcomes	Intervention Percentage	Comparator Percentage	Relative effect (95% CI)	Anticipated absolute effects (95% CI) Difference	Certainty of the evidence (GRADE)
Death (all-cause) N=1145	1.2%	2.3%	RR 0.54 (0.22 to 1.34)	10 fewer per 1,000 (from 18 fewer to 8 more)	⊕⊕⊕○ Moderate
Treatment success N=1145	96.9%	96.9%	RR 1.00 (0.98 to 1.02)	0 fewer per 1,000 (from 19 fewer to 19 more)	⊕⊕⊕⊕ High

Treatment failure N=1145	0.50%	0.20%	RR 3.01 (0.31 to 28.81)	4 more per 1,000 (from 1 fewer to 49 more)	⊕⊕⊕⊕ High
Relapse N=1145	1.0%	0.7%	RR 1.50 (0.43 to 5.30)	3 more per 1,000 (from 4 fewer to 30 more)	⊕⊕⊕○ Moderate
Treatment adherence N=1204	95.1%	93.2%	RR 1.02 (0.99 to 1.05)	19 more per 1,000 (from 9 fewer to 47 more)	⊕⊕⊕○ Moderate
Adverse events during treatment (grade 3 or higher) N=1204	7.8%	8.0%	RR 0.98 (0.67 to 1.44)	2 fewer per 1,000 (from 26 fewer to 35 more)	⊕⊕⊕○ Moderate
Loss to follow up N=1204	1.8%	1.8%	RR 1.00 (0.44 to 2.29)	0 fewer per 1,000 (from 10 fewer to 24 more)	⊕⊕⊕○ Moderate

* Adapted from WHO DS 2022 TB treatment guidelines Evidence to Decision tables (From EtD corresponding to question “A 4-month treatment regimen compared to currently recommended 6-month treatment regimen in children and adolescents with non-severe drug-susceptible tuberculosis”, pages 45-58. These numbers and terms may differ from other publications of the underlying studies due to differences from specific WHO related definitions, assessment and analysis methodology of the primary data.

† Interventional regimen of 8-week intensive phase of daily fixed-dose combination of isoniazid, rifampin, pyrazinamide with inclusion/exclusion of ethambutol per local guidelines, followed by 8-week continuation phase of daily isoniazid and rifampin

‡ Control regimen of 8 week intensive phase of daily fixed-dose combination of isoniazid, rifampin, pyrazinamide with inclusion/exclusion of ethambutol per local guidelines, followed by 16-week continuation phase of daily isoniazid and rifampin

Table 7. Subgroup considerations for the 4-month 2HRZE/2HR regimen for children with non-severe DS-TB

Subgroup	SHINE study eligibility and outcomes	Management Considerations/Recommendations
Children with asymptomatic disease	Not study eligible.	Asymptomatic children with non-severe TB disease should receive the 4-month recommended regimen even without microbiologic results
Children with peripheral lymph node TB	Included in study. Despite small numbers, the 4-month regimen was non-inferior for peripheral lymph node disease across all subgroups, including those with concomitant pulmonary disease.	Peripheral and intrathoracic lymph nodes may remain enlarged and regress over several months We recommend the 4-month regimen.
Children living with HIV	Included in study. The 4-month regimen was non-inferior regardless of antiretroviral treatment and CD4 count, including those with severe HIV infection [per WHO immunological classification].	Monitor children with HIV on this regimen closely, especially with severe HIV disease or opportunistic infections, and extend to 6 months if insufficient clinical progress at 4 months. We recommend the 4-month regimen.
Children with severe acute malnutrition [SAM]	Included in study, but no separate subgroup analysis. In SHINE trial, SAM was defined as weight-for-height Z-score ≤ 3 or mid-upper arm circumference of <115 mm.	As no separate subgroup analysis conducted, children with SAM and non-severe disease should preferably receive 6-month regimen
Premature Infants < 37 weeks' gestation and aged < 3 months Or Infants weighing < 3 kg but ≥ 37 weeks gestation	Not study eligible.	These children should be treated with a 6-month regimen
Children treated for TB in the past 2 years	Not study eligible.	Generally not eligible for this regimen, but consider individual circumstances and assess if appropriate for the 4-month regimen.
Children exposed to drug-resistant TB (DR-TB)	Not study eligible	These children should be treated with a regimen appropriate for the DR-TB exposure

Table 8. TB-PRACTECAL, ZeNIX trial evidence: Drug regimens and WHO-analyzed critical outcomes in the treatment of rifampin-resistant, fluoroquinolone resistant/sensitive TB*

Study	TB PRACTECAL *, †, §,	ZeNix*, ‡, §, , **	WHO Long SoC Regimens Registries: Individual Patient Dataset (IPD) 2021*
Design	Phase 2/3 open-label RCT, two stages with randomization to: I. BPaL or WHO Long Standard of Care II. BPaLM or WHO Long Standard of Care	Open-label RCT: BPaL 600 for 9 or 26 weeks, or BPaL 1200 for 9 or 26 weeks	Program registry observational data
Setting	Belarus, South Africa, Uzbekistan	South Africa, Georgia, Moldova, Russia	Belarus, Georgia, Moldova, Somalia, India, Mozambique, Belgium, Armenia, South Africa, and multiple sites from Einstein, End TB, CDC, Harvard and France datasets.
Inclusion Criteria	-MDR/RR-TB or -pre-XDR-TB -above regardless of quinolone resistance or HIV status -aged 15 years or more	-XDR TB, -pre-XDR, -RR TB -aged 14 years or more	For long comparator regimen (target 18–24 months), patients had each of the following: -Regimen classified as long regimen in dataset -Treatment duration ≤ 24 months -Received ≥ 4 drugs (regardless of DST or effectiveness) - bedaquiline allowed in regimen -If cure or completion, treatment duration of 17.5 months or more.

Exclusion Criteria	-Pregnancy, -hepatic or heart disease, -suspected resistance to bedaquiline, pretomanid, or linezolid - prior use of bedaquiline and/or pretomanid and/or linezolid for one or more months. HIV+ & CD4<100, ALT and AST > 3X ULN, Grade >3 peripheral neuropathy, prior treatment with any of 3 trial drugs or delamanid for 2+ weeks Patients receiving injectable antibiotics excluded				
Regimen (see footnotes)	TB PRACTECAL BPaL n=126, *†§	TB PRACTECAL BPaLM n=128, * † II	WHO Long SoC, with injectables*	ZeNix BPaL (600mg) *‡ **	WHO Long SOC, Individual Patient Data registry
Duration	24 weeks	24 weeks	9-20 months	26 weeks	variable
WHO Outcomes*	%	RR †† (95% CI)	%	RR†† (95% CI)	%
Success rate	76.7	1.47 (1.09 to 1.99)	88.7	1.73 (1.31 to 2.27)	51.5
Failure and recurrence	13.3	0.52 (0.22 to 1.18)	8.1	0.26 (0.10 to 0.71)	25.8
Loss to Follow-up	10.0	0.60 (0.24 to 1.56)	3.2	0.16 (0.04 to 0.61)	19.7
				RD †† - 0.11 (-0.13 to -0.03)	11.8

Adverse events	19.6	0.38 (0.24 to 0.60)	21.0	0.41 (0.26 to 0.63)	50.9	14.0	RD†† - 0.12 (-0.14 to -0.04)	4.7
Death	0.0	RD ‡‡ 0.03 (-0.1 to 0.03)	0.0	RD‡‡ - 0.03 (-0.1 to 0.03)	3.0	0.0	3.99 (1.67 to 9.57)	11.0
Amplified resistance	2.9	1.59 (0.32 to 7.84)	0.0	RD‡‡ - 0.02 (-0.07 to 0.02)	1.9	0.0	RD ‡‡ - 0.02 (- 0.04 to 0.06)	2.4

* Adapted from WHO DS 2022 TB treatment guidelines evidence tables and Evidence to Decision tables, specifically corresponding to WHO questions “Should BPaL (Lzd 600mg/300mg) vs. local SoC regimens (TB-PRACTECAL comparator) be used for pulmonary MDR/RR-TB and pre-XDR-TB? And question “and “Should BPaL vs. WHO long be used for pulmonary MDR/RR-TB? “, the data for the percentages of participant for each outcome and the treatment effect estimates were derived from the evidence tables and Evidence (EtD) to decision respectively (Module 4 Treatment, Web annex under PICO 6.6 pages 47-48 [evidence table] and 294-309 [EtD] and PICO 5.2 pages 33-34 [evidence table] and 196-212[EtD]). Additionally, for question “BPaLM compared to TB-PRACTECAL comparator in pulmonary MDR/RR-TB and pre-XDR-TB.” The data for the percentages of participant for each outcome and the treatment effect estimates were derived from the evidence tables and Evidence (EtD) to decision respectively. (Module 4 Treatment, Web annex under PICO 6.1 pages 37-38 [evidence table] and 226-241[EtD]). These numbers and terms may differ from other publications of the underlying studies due to differences from specific WHO related definitions, assessment and analysis methodology of the primary data.

† Compared to WHO Long Standard of Care (SoC)

‡ Compared to WHO long Individual Patient Data registry

§ TB-PRACTECAL BPaL: bedaquiline 400 mg daily for 2 weeks then 200mg 3 times a week for 22 weeks; pretomanid 200 mg daily for 24 weeks; linezolid 600 mg daily for 16 weeks, then 300 mg daily for 8 weeks.

|| TB-PRACTECAL BPaLM: bedaquiline 400 mg daily for 2 weeks then 200mg 3 times a week for 22 weeks; pretomanid 200 mg daily for 24 weeks; linezolid 600 mg daily for 16 weeks, then 300 mg daily for 8 weeks; moxifloxacin 400mg daily for 24 weeks

** ZeNix BPaL (600mg): ZeNix (BPaL with 600mg linezolid daily for 26 weeks).

†† Risk Ratio

‡‡ Real Difference (RD) is used for comparison instead of RR since numerator for a risk requires a non-zero number

Table 9. Monitoring plan for patients treated with BPAL or BPALM*

Activity	Month of Treatment											Post-treatment ¹⁸				
	0 (Baseline)	1 +		2	3	4	5	6	7	8	9	3	6	12	18	24
Sputum smear and culture ¹	●	●	●	●	●	●	●	●	○	○	○	●	●	●	●	●
Imaging (CXR, CT, other) ²	●				●			●			○	○	○	○	○	○
Weight ³	●	●		●	●	●	●	●	○	○	○	●	●	●	●	●
Symptom review ⁴	●	●		●	●	●	●	●	○	○	○	●	●	●	●	●
DST ⁵	●				○											
CBC ⁶	●	●	●	●	●	●	●	●	○	○	○					
Creatinine ⁷	●	●		●	●	●	●	●	○	○	○					
ALT/AST, alkaline phosphatase, bilirubin ⁸	●	●		●	●	●	●	●	○	○	○					
K+, Ca++, Mg++, bicarbonate ⁹	●	●		●	●	●	●	●	○	○	○					
Serum drug concentration ¹⁰		○														
HIV ¹¹	●															
Pregnancy ¹²	○	○		○	○	○	○	○	○	○	○					
EKG ¹³	●	●	○	○	●	○	○	●	○	○	○					
Vision Exam ¹⁴	●	●		●	●	●	●	●	○	○	○					
Peripheral Neuropathy ¹⁵	●	●		●	●	●	●	●	○	○	○					
Arthralgias ¹⁶	●	○		○	○	○	○	○	○	○	○					
Amylase, lipase, TSH ¹	○															

*Joint Panel opinion, adapted from California MDR Service.

† Split column indicates weekly or biweekly testing.

● Indicates activity is recommended by Joint Panel opinion

○ Indicates activity that is optional or contingent on other information.

Monitoring recommendations may change if treatment regimen or patient status changes.

When treatment extended to 9 months, monitoring for month 7-9 (weeks 27-39) would be "as needed."

1. Obtain 3 sputa at the start of treatment and every 1-2 weeks until smear conversion, followed by 1-2 sputa every 2 weeks until culture conversion, and then 1 sputum monthly throughout treatment.
2. Obtain baseline imaging and monitor approximately every 3 months until end of treatment.
3. Monitor weight monthly and adjust medications as needed.
4. Monitor for TB symptoms monthly.
5. Obtain 1st and 2nd-line DSTs at baseline. Repeat if culture remains positive after 2-3 months of treatment.
6. Obtain CBC every 1-2 weeks for the first 6-8 weeks, then monthly while on LZD. A decrease of hemoglobin of greater than 10% or more at 4 weeks of treatment may identify those at high risk for severe anemia.
7. Obtain serum creatinine at baseline and monthly while on BPaL or BPaLM.
8. Obtain ALT, AST, alkaline phosphatase, bilirubin at baseline and then monthly while on BDQ and Pa.
9. Obtain K⁺, Ca⁺⁺, Mg⁺⁺, and bicarbonate at baseline and monthly while on BDQ.
10. Obtain therapeutic drug monitoring (TDM), if possible, of LZD after 1-2 weeks on therapy and if signs of toxicity develop. Recommend collecting a LZD trough concentration if resources allow. TDM may be obtained for other drugs as clinically indicated.
11. Obtain baseline HIV test.
12. Consider baseline and monthly pregnancy test for patients capable of becoming pregnant who decline long-acting forms of birth control (e.g., intrauterine devices or implantable contraception).
13. Obtain EKG (check QTcF) at baseline, 2, 12, and 24 weeks while on BDQ. Consider checking monthly if taking additional QT-prolonging agents (e.g., MFX, CFZ, other).
14. Perform visual acuity (e.g., Snellen) + color discrimination (i.e., Ishihara) exams at baseline and monthly while on LZD.
15. Monitor for peripheral neuropathy at baseline and monthly while on LZD.
16. Monitor for arthralgias at baseline and monthly while patient on a FQ.
17. Note that CDC provisional guidance (ref. 3) for use of BPaL recommends amylase, lipase, and TSH at baseline. These may not be necessary for all patients, according to Joint Panel opinion. Consider checking amylase and lipase if underlying concerns for pancreatitis or if symptoms develop. Consider checking TSH if there are concerns for prolonged QT interval on baseline EKG.
18. Recommend clinical assessment (including physical exam and symptom review), sputum collection (2-3 specimens), and chest imaging at 3, 6, 12, 18, and 24-months post-treatment completion.

Table 10. Research needs

<i>Diagnostic and translational studies</i>
Expansion of methods and use of culture-free rapid DST for moxifloxacin
Description of the mechanism and molecular markers of pretomanid resistance, allowing development of DST methods
Analyses of cross-resistance between drugs in current regimens with delamanid and surveillance for the development of resistance
Analyses of cross-resistance between bedaquiline and clofazimine and surveillance for the development of resistance
Assessment of the feasibility and accuracy of applying definitions for non-severe TB in routine clinical practice with particular emphasis on the radiographic components of the definition for children and adolescents <16 years old
<i>Treatment and outcomes</i>
Evaluation of longer-term outcomes for newly recommended regimens
Reports of efficacy, safety, and tolerability of newly recommended regimens for subpopulations for whom current data are limited or missing (pregnant and lactating people, young children, older adults, people with significant renal or hepatic disease)
Evaluation of these regimens for treatment of extra-pulmonary and disseminated forms of tuberculosis
Evaluation of treatment data from other regions and countries using these regimens
Assessment of barriers to implementation of shorter regimens
Evaluation of the uptake of shortened treatment regimens
Evaluation of the acceptability of shorter treatment regimens from patient, medical provider, and other perspectives
Evaluation of efficacy and added value in multidrug regimens of pretomanid and delamanid

Determination of efficacy of other and even shorter regimens
Evaluation of outcomes for which current evidence is scarce (e.g., acquisition of drug resistance and quality of life)
Examination of the geographical differences in the frequency and severity of linezolid-related adverse events and the underlying cause
Testing replacement of moxifloxacin with levofloxacin in newly recommended regimens
Defining the role of and methods for ECG monitoring with newly recommended regimens, particularly for older adults
Implementation of antibiotic stewardship measures for TB and community use of fluoroquinolones
Use of pharmacokinetic data to refine dosing or for use in treatment monitoring
Identification of strategies to ensure treatment adherence and completion to cure
Data collection for cost-effectiveness and feasibility of newly recommended regimens
Impact on health equity and acceptability of newly recommended regimens
Ascertainment of efficacy of regimens by TB lineage
Evaluating efficacy of adding clofazimine to BPaL in fluoroquinolone-resistant TB
Analyses of pharmacokinetic data for crushed tablets used in treating children with TB
Developing child friendly formulations for TB medications in high income countries
Economic analyses of shorter regimen costs, DST, and monitoring

Abbreviations:

BPaL- Bedaquiline, pretomanid, and linezolid

DST- Drug susceptibility testing

TB- Tuberculosis